

EXPERIMENT: 5

- i. Create a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found).
- ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block..

i). We have to create the student table and insert the records in to the table as follows:

```
SQL> create table student(sid number(10),sname varchar2(20),rank varchar(10));
```

Table created.

```
SQL> insert into student values(501,'Ravi','second');
```

1 row created.

```
SQL> insert into student values(502,'Raju','third');
```

1 row created.

```
SQL> insert into student values(503,'Ramu','');
```

1 row created.

```
SQL> select *from student;
```

```
SID SNAME RANK
```

```
501 Ravi second
```

```
502 Raju third
```

```
503 Ramu
```

PL/SQL CODE:

```
SQL>ed 5a
```

Enter the following code into the text editor and save the file with .sql format

```
set serveroutput on;
```

```
declare
```

```
    temp1 number(10);
```

```
    temp2 varchar2(10);
```

```
begin
```

```
    select sid,sname into temp1,temp2 from student where rank='first';
```

```

        dbms_output.put_line('Student No:'|| temp1 ||' Name:'||temp2||' got first rank');
exception

        when no_data_found then
        dbms_output.put_line('*****');
        dbms_output.put_line('# Error: there is no student got first rank');

end;

/

```

SQL> @5a;

Error: there is no student got first rank

PL/SQL procedure successfully completed.

SQL> update student set rank='first' where sid=503;

1 row updated.

SQL> select *from student;

SID SNAME RANK

501 Ravi second

502 Raju third

503 Ramu first

SQL> @5a

Student No:503 Name:Ramu got first rank

PL/SQL procedure successfully completed.

ii) SQL> select *from student;

SID SNAME RANK

501 Ravi second

502 Raju third

503 Ramu first

PL/SQL CODE:

SQL>ed 5b

Enter the following code into the text editor and save the file with .sql format

```

set serveroutput on;

DECLARE

    sno student.sid%type;

    name student.sname%type;

    srnk student.rank%type;

BEGIN

    sno := &sno;

    name := '&name';

    srnk := '&srnk';

    INSERT into student values(sno,name,srnk);

    dbms_output.put_line('One record inserted');

    COMMIT;

    -- adding savepoint

    SAVEPOINT s1;

    -- second time asking user for input

    sno := &sno;

    name := '&name';

    srnk := '&srnk';

    INSERT into student values(sno,name,srnk);

    dbms_output.put_line('One record inserted');

    ROLLBACK TO SAVEPOINT s1;

END;

```

/

SQL> @5b;

Enter value for sno: 504

old 7: sno := &sno;

new 7: sno := 504;

Enter value for name: ali

old 8: name := '&name';

new 8: name := 'ali';

Enter value for srnk: first

old 9: srnk := '&srnk';

new 9: srnk := 'first';

Enter value for sno: 505

old 16: sno := &sno;

new 16: sno := 505;

Enter value for name: haji

old 17: name := '&name';

new 17: name := 'haji';

Enter value for srnk:

third old 18: srnk := '&srnk';

new 18: srnk := 'third';

One record inserted

One record inserted

PL/SQL procedure successfully completed.

SQL> select *from student;

SID SNAME RANK

501 Ravi second

502 Raju third

503 Ramu first

504 ali first